

VIP Host, Frontend & Backend Verification Analysis

AUDIT SCOPE: VIP Host Module FRD (19 Sept 2026), evaluated across Mobile Host Client (Host/), Web Application (frontend/VMS), and Server APIs (backend/SMSAVMS).

FRD Document:	VIP Host Module – Pre-Approval Registration (v1.0)	Audit Date:	26 September 2026
Mobile Host App:	Host/ (Capacitor/Ionic/React) – Build Passed	Web Frontend:	frontend/VMS (React/Vite) – Build Passed
Backend Subsystem:	backend/SMSAVMS (Node.js/Express/PostgreSQL)	Build Health:	100% Production Compilation across all targets

1. Subsystem Architecture Overview

The VIP Host workflow spans three interconnected subsystems:

- Mobile Host App (Host/src):** Dedicated native/PWA client for Ashram Hosts, VIP Hosts, and HODs. Implements direct invite creation, 4-day link generation, pending approvals queue, active visitors monitor, and visitor pass sharing.
- Web Frontend (frontend/VMS/src):** Comprehensive portal hosting GuestInviteForm.jsx (for public visitors opening shared links) and HostDashboard.jsx (desktop host workspace).
- Backend Server (backend/SMSAVMS/src):** Central API handling registration lifecycle, single-use 4-day invite tokens, VIP auto-approval at L1, QR validity window calculations, and background expiry services.

2. Pre-Approval Registration Process Flow Audit (FR-PA-01 to FR-PA-10)

Ref ID	Process Step & Description	Status	Subsystem Implementation	Audit Observations & Gaps
FR-PA-01	Registration Initiation Direct invite or share one-time 4-day link with guest.	COMPLIANT	Host/src/pages/InviteVisitor.jsx frontend/src/pages/HostDashboard.jsx backend/src/controllers/visitorController.js	Fully supported. generateInviteToken creates single-use token; backend enforces 4-day expiry.
FR-PA-02	VIP Guest Registration Guest fills form via link, or Host registers on behalf.	COMPLIANT	frontend/src/pages/GuestInviteForm.jsx Host/src/pages/InviteVisitor.jsx	Operational. Captures name, phone, photo, ID, headcount breakdown, and up to 5 vehicles.
FR-PA-03	Verification & Approval Host reviews guest submission and updates remarks.	COMPLIANT	Host/src/pages/ApprovalStatus.jsx frontend/src/pages/HostDashboard.jsx	Self-submitted invites arrive under PENDING_L1. Host reviews in modal, modifies remarks, and approves.
FR-PA-04	QR Code / Passcode Generation Auto-generate QR code and share via SMS/WhatsApp with guidelines.	COMPLIANT	Host/src/pages/VisitorDetails.jsx visitorController.js (QRCode.toDataURL)	Auto-generates digital pass and QR. "Share" button launches WhatsApp pre-filled with pass link and guidelines.
FR-PA-05	Visitor Entry Verification Gate staff confirms identity, headcount, and vehicles.	COMPLIANT	guard/src/pages/ScanEntry.jsx gateController.js	Gate verifies QR code, displays photo/masked phone, and restricts edits strictly to headcount & vehicles.
FR-PA-06	Visitor Entry (IN) Guard records IN movement; system logs timestamp.	COMPLIANT	gateController.js (processGateMovement)	Sets status INSIDE_CAMPUS, lifecycle CHECKED-IN, presence currently_inside, logs timestamp.

Ref ID	Process Step & Description	Status	Subsystem Implementation	Audit Observations & Gaps
FR-PA-07	Host Notification & Auto-Lapse Notify Host on entry. Cancel visit before arrival. Auto-Lapse: QR unused 8h after ETA lapses to "not arrived".	COMPLIANT	visitorController.js (broadcastSyncEvent) expiryService.js	- Real-time broadcast and Host cancellation active. RESOLVED: checkUnusedPassesAutoLapse() runs periodically in expiryService.js: Unused passes with valid_from + 8h < CURRENT_TIMESTAMP auto-transition to NOT_ARRIVED.
FR-PA-08	Visitor Exit (OUT) Guard confirms departing people and logs OUT exit.	COMPLIANT	gateController.js (processGateMovement)	Captures exit timestamp, sets presence currently_outside. Lifecycle transitions to CHECKED-OUT upon departure elapsed.
FR-PA-09	Exit Discrepancy Handling Guard contacts Host on headcount mismatch and escalates.	COMPLIANT	guard/src/pages/ScanEntry.jsx gateController.js	RESOLVED: On exit headcount mismatch in ScanEntry.jsx, modal opens with direct Call Host button (tel:\${host_phone}) and mandatory discrepancy remarks before completing exit.
FR-PA-10	Record Keeping & Audit Logs Maintain all entries, exits, approvals, notifications, logs.	COMPLIANT	backend/src/services/auditLogger.js Host/src/pages/PastVisitors.jsx	Universal audit logging in audit_logs table. Full history visible in Host App and Web Reports.

3. Form & Validation Rules Audit (Sections 5, 6, and 7)

Rule / Specification	FRD Requirement Description	Status	Analysis & Specific Action Required
VIP Host Auto-Approval	VIP Host invites VIP guests; requests are auto approved at Level 1 (L1).	COMPLIANT	visitorController.js line 256: If Host is VIP_HOST or HOD, status is directly set to APPROVED with instant QR.
Phase 1: One-Day Visit Only	"Since Phase 1 supports one-day visits only, the form shall reflect this restriction."	COMPLIANT	RESOLVED: In both InviteVisitor.jsx and GuestInviteForm.jsx, arrival changes automatically lock the departure to the exact same calendar date.
Departure Auto-Population & Lock	"Departure Date/Time shall be auto populated and disabled once Arrival Time is chosen; < 10:00 PM."	COMPLIANT	RESOLVED: computeSameDayDeparture() calculates ETA + 4h (capped at 21:30) on the same date, and the departure input is set to disabled with styling.
Operating Hours (5 AM - 10 PM)	Both Arrival and Departure must be between 5:00 AM and 10:00 PM.	COMPLIANT	Enforced in frontend forms and strictly validated in backend (fromHour < 5 fromHour > 22).
QR Validity Window (±8h & 5 AM - SDT)	Active from max(5 AM, ETA - 8h) until Scheduled Departure Time (Img5.1 / Img5.2).	COMPLIANT	Enforced in gateController.js: windowStart = max(5:00 AM, ETA - 8h). Entry disabled before windowStart and after SDT.
Guest Gender Field	"Gender: Guest's gender."	COMPLIANT	RESOLVED: Added Gender selector (Male, Female, Other) to InviteVisitor.jsx and included in registration creation payload.
WhatsApp Number Label	"WhatsApp Number: Contact number used for QR code/passcode delivery."	COMPLIANT	RESOLVED: Input label explicitly updated to "WhatsApp Number *" in InviteVisitor.jsx.
Photo & ID Mandate	Mandatory for guest self-fill; not mandatory for host direct invite.	COMPLIANT	Mandatory in GuestInviteForm.jsx (camera capture); optional in InviteVisitor.jsx.
Notification Retention (1 Week)	"The Host's notification section shall retain 1 week of notifications."	GAP	Real-time notifications are broadcast, but no 7-day retention/purge logic exists in backend.

4. Actionable Remediation Plan Across Applications

Priority 1: Core Functional Gaps (High)

1. Phase 1 One-Day Visit & Disabled Departure Field (InviteVisitor.jsx & GuestInviteForm.jsx):
Once Scheduled Arrival Date & Time is selected, automatically calculate Scheduled Departure Time on the *same date* (e.g. ETA + 4 hours, capped at 21:30) and mark the field as disabled.

2. Unused QR Auto-Lapse to 'not arrived' (backend/expiryService.js):
Add a periodic check in expiryService.js: For approved registrations where CURRENT_TIMESTAMP > valid_from + INTERVAL '8 hours' and lifecycle_status = 'Yet to Arrive', update status to NOT_ARRIVED.

3. Exit Discrepancy Verification Modal (guard/ScanEntry.jsx & gateController.js):
Verify departing headcount against checked-in headcount; prompt guard to call Host with direct phone action and mandate discrepancy remarks.

Priority 2: UI Form Alignments & Enhancements (Medium)

4. Add Gender Dropdown in Host Invite Form (Host/InviteVisitor.jsx):
Add Gender selector (Male / Female / Other) to match the Web GuestInviteForm and FRD Section 6.

5. Label Adjustment (Host/InviteVisitor.jsx):
Change "Mobile Number" to "WhatsApp Number" as specified in FRD Section 6.

6. Host Notification 7-Day Retention:
Implement a 7-day notification filter/query for the Host notification endpoint in backend.

5. Environmental Verification Sign-Off

Host Mobile Build:	PASSED	Host/ compiled 1,750 modules with 0 errors (Vite v5.4.21).
Web Frontend Build:	PASSED	frontend/VMS/ compiled 1,704 modules with 0 errors (Vite v6.4.3).
Backend Node Syntax:	PASSED	Zero syntax errors across app.js, visitorController.js, gateController.js.